GENERATIVE AI FOR SCIENCE

A Hands-On Guide for Students and Researchers

PROF. J. PAUL LIU

Generative AI for Science

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Contents

From the Author

A Journey from Classroom to Lab — and Now to You

This book is the result of years of teaching, research, and hands-on collaboration at the intersection of generative AI and scientific discovery. Everything here has been shaped by real use: tested in graduate courses, refined in workshops, improved through research projects, and validated by hundreds of scientists applying these tools to their own work.

Origins

The material began in my graduate course **Generative AI for Science**, taught at the Data Science and AI Academy, where students from biology, chemistry, physics, materials science, and computational fields explored a fundamental question:

How can AI not only analyze data, but generate new scientific knowledge?

Lecture notes grew into full tutorials as we worked through real challenges: predicting protein stability, designing new materials, accelerating climate simulations, and mining insights from vast scientific corpora. Student questions sharpened the explanations; their projects highlighted what worked; their struggles revealed where clarity was needed.

Beyond the classroom, the methods in this book have been strengthened through:

- Bioinformatics Research Center workshops
- Cross-campus **AI for Research** training programs
- Research Triangle AI Society–LLM intensive bootcamps
- Active collaborations in oceanography, materials science, protein engineering, and literature mining

What Makes This Book Different

This book is *not* a traditional AI textbook. Rather than focusing on heavy theory, it provides a hands-on, practical guide to using generative AI in real scientific workflows through educational case studies. It is ideal for college students, researchers, and data scientists who want to learn by doing.

- **Theory meets practice:** Every concept is paired with ready-to-run code.

- **Interactive learning:** All example codes are provided as Google Colab notebooks—no installation required.
- **Real scientific problems:** Examples come from authentic research.
- **Accessible yet rigorous:** Suitable for both domain scientists exploring AI and ML experts entering scientific applications.

How to Use This Book

As a course text:
Follow the chapters sequentially for a structured introduction.

As a reference:
Jump directly to the sections relevant to your research domain, or follow the further reading references to explore topics in greater detail.

As a hands-on guide:
Open the Colab notebooks or Powerpoint slides alongside each chapter, run the code, and modify it as you learn.

As a research launchpad:
Use the provided implementations as starting points for your own projects.

What You Will Learn

By the end of this book, you will:

- Understand key AI architectures such as Transformers, Diffusion Models, VAEs, and GNNs
- Represent scientific data types effectively
- Apply generative models to problems in climate science, drug discovery, genomics, materials science, and more
- Follow best practices around ethics, reproducibility, and deployment
- Stay current with emerging methods and future directions

Most importantly, you will develop the intuition to know *when* and *how* to apply AI to scientific research.

A Note on Collaboration

AI-accelerated science thrives at the intersection of domain expertise and computational skill. Biologists will gain enough ML understanding to explore confidently, while ML practitioners will gain enough scientific context to ask meaningful questions and validate results. This book is itself a perfect example of how I collaborate with AI on writing, coding, editing, and proofreading.

Collaboration is the catalyst. This book aims to provide the shared language needed to bridge fields and perspectives.

Let's Begin

Generative AI does not replace the scientific method—it enhances it. It expands the space of hypotheses we can explore, sharpens experimental design, and reveals patterns hidden in complexity. But the foundations remain unchanged: rigor, honesty, and careful interpretation.

Combine human creativity with machine assistance, and new discoveries become possible.

Now, let us explore them together.

Dr. J. Paul Liu December 2025

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Downloading all codes or slides

Access all Colab notebooks and slides: https://github.com/jpliu168/Generative_AI_For_Science

Prerequisites

- Basic Python programming (functions, loops, data structures)
- Undergraduate-level statistics (distributions, hypothesis testing)
- Familiarity with scientific computing (NumPy, Matplotlib helpful but not required)
- No prior deep learning experience necessary

Technical requirements: A web browser and curiosity. Everything else runs in the cloud.

AI Use Disclaimer

This work was developed with the assistance of modern AI language models, including **ChatGPT**, **Claude**, and **Gemini**, which supported:

1. Improving clarity of writing, including corrections to English grammar and syntax
2. Text organization and formatting, including Markdown structure, embedded Python code blocks, output results, and visual styling
3. Code generation, formatting, debugging, and proofreading

Writing Workflow

Here is the workflow I followed for most chapters:

1. **Research and preparation:** Based on my previous class notes, sample codes, and PowerPoint slides, I first conducted extensive Google searches and read through hundreds of links, papers, and reports.
2. **Initial AI consultation:** I posed my questions, ideas, and plans to AI assistants. With their suggestions, I refined my questions and ideas, returned to Google searches and reference readings, and then asked AI for help with the book proposal.
3. **Drafting:** For each chapter, I sent AI the topic, questions, my references, and sample codes to generate an initial draft. I then manually edited and updated the content by combining my own data, new references, and updated codes.
4. **Code development:** I spent the majority of my time coding, debugging, customizing, and finalizing each case study to ensure it could run within typical student computational resources for educational demonstration purposes.
5. **Validation:** I shared some chapters with students and colleagues for their input. Based on their feedback, I revised and updated the content further.
6. **Final polish:** AI assisted in polishing all my writing and reformatting each section to fit the Leanpub Markua format.

Many times, to meet this book's educational demo purpose, I had to compare, adjust, or combine results from different LLMs based on the quality of their responses. I retain full responsibility for all conceptual content, workflows, research interpretations, citations, discussions, conclusions, and final decisions.

Through this intensive collaborative writing and coding process, I learned a great deal and cannot wait to share all of this in my next class offering of *Generative AI for Science*.

Chapter 1 — Generative AI: A New Frontier for Scientific Discovery

The New Frontier of Scientific Discovery

We stand at an extraordinary moment in the history of science. For centuries, the pace of discovery has been limited by human capacity—our ability to read literature, design experiments, analyze data, and synthesize knowledge. But a fundamental shift is underway. **Generative Artificial Intelligence (AI)** is not merely automating workflows; it is expanding what is scientifically possible [1].

Just as the microscope, the computer, and genome sequencing each transformed how we perceive nature, Generative AI now transforms our ability to *understand* and *create* scientific knowledge. AI systems design novel antibiotics by exploring chemical spaces that would take humans millennia to search [2, 3]. They predict protein structures with atomic precision—work recognized with the 2024 Nobel Prize in Chemistry [4, 5]. AI co-scientists generate hypotheses from vast textual corpora in days that previously required years of iterative research [6, 7], identify patterns in climate and Earth system data that are difficult to capture with traditional methods [8–10], and propose entirely new materials and reaction pathways [3, 11].

This is not science fiction—this is the reality of Generative AI today [1].

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The AI Revolution in Scientific Discovery

From Analysis to Generation

Scientific computing has evolved through clear stages:

Era	Core Capability	Tools	Outcome
1960s–1980s	Digitization	Databases	Data preservation
1990s–2010s	Statistical & ML methods	Regression, SVMs, random forests	Pattern recognition

Era	Core Capability	Tools	Outcome
2020s–	Generative AI	Transformers, diffusion models	Creation of new data & hypotheses

Traditional scientific discovery has often followed a cycle of *observe* → *hypothesize* → *experiment* → *analyze* → *repeat* [12]. Generative AI compresses and accelerates this loop [1, 7]:

- **Literature review:** Synthesizes insights from millions of papers [6, 7].
- **Hypothesis generation:** Identifies non-obvious links across fields, with AI co-scientists reducing hypothesis generation from weeks to days [1, 7].
- **Experimental design:** Proposes thousands of candidate molecules or materials for testing [2, 3, 11].
- **Data analysis:** Finds structure in high-dimensional scientific datasets [8–10, 12].

The Acceleration Effect

The effect is both quantitative and qualitative [1]:

- **Speed:** AI approaches can shorten drug discovery timelines from ~10–15 years to ~3–5 in some workflows, with AI-designed drugs showing 80–90% success rates in Phase I trials compared to traditional 40–65% [2, 13].
- **Scale:** Virtual screening has been used to evaluate extensive compound libraries in silico for drug discovery, supporting efficient prioritization of candidates[3].
- **Efficiency:** Weather forecasts that once required hours on supercomputers can be generated in minutes with AI models like GenCast [9, 14].

More profoundly, AI enables scientists to explore vast combinatorial spaces and ask *what if?* at scales that would be infeasible with manual or purely equation-based approaches [1, 12].

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What Makes Generative AI Different from Traditional ML?

Traditional Machine Learning — Pattern Recognition

Traditional machine learning (ML) excels at **discriminative** tasks such as classification, regression, and clustering—learning mappings from inputs to outputs and identifying patterns in existing data [12].

Generative AI — Creating the New

Generative models go further: they learn the **distribution** underlying the data and can sample from it to create new, realistic instances [15, 16]. This enables capabilities such as:

Capability	Example	Outcome
Generation	Design a new molecule with desired properties	Novel compounds [2, 3]
Completion	Fill missing regions in a protein structure or sequence	Plausible biological variants [4, 5]
Translation	Text description → molecule, code, or figure	Cross-modal synthesis [1, 6]
Synthesis	Combine text, code, and data	New simulations, analyses, or visualizations [6]

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Core Technologies Powering Generative AI

1. Transformers and Large Language Models (LLMs)

Transformers introduced an attention-based architecture that models long-range dependencies and contextual relationships in sequences [15]. Large Language Models (LLMs) built on this architecture can:

- Summarize and synthesize scientific literature
- Generate runnable analysis code and scripts
- Act as domain-aware assistants that interact with text, equations, and data [1, 6]
- Serve as AI co-scientists that generate novel hypotheses through multi-agent debate and evolution [7]

2. Diffusion Models

Diffusion models learn to reverse a gradual noising process to generate high-fidelity samples [16–18]. They are increasingly used for:

- Molecular and materials design [3, 11]
- Protein and structural biology tasks, including AlphaFold 3's prediction of biomolecular interactions [4, 5]
- Probabilistic weather forecasting with models like GenCast [9, 14]

3. Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs)

Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) provide earlier but still important frameworks for generative modeling [19, 20]. They support:

- Learning low-dimensional latent representations of complex systems
- Creating synthetic datasets to augment limited experimental data
- Constructing surrogate models for expensive simulations

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The Pre-Training Revolution

Modern generative AI thrives on **pre-training**: models are first trained on large, generic corpora (text, code, images, or scientific data) and then **fine-tuned** on specific tasks or domains [6]. This strategy is particularly valuable in science, where high-quality labeled data are often scarce or costly to obtain.

A language model pre-trained on broad text can be adapted to domains such as biomedicine, materials science, or geosciences. Protein models pre-trained on millions of sequences can specialize in a narrow protein family with comparatively small task-specific datasets [4, 5].

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Generative AI Across Scientific Disciplines

Physical and Chemical Sciences

In the physical and chemical sciences, generative AI is already reshaping how we search, design, and reason about complex systems:

- **Materials Design:** Exploring enormous chemical spaces to propose new superconductors, catalysts, or battery materials with targeted properties [3, 11].
- **Reaction and Synthesis Planning:** Suggesting reaction products and retrosynthetic routes, accelerating the path from concept to synthesis [3].
- **Mathematical Reasoning:** AlphaProof and AlphaGeometry 2 achieved silver-medal level performance at the 2024 International Mathematical Olympiad, and an advanced Gemini model achieved gold-medal standard in 2025 [21, 22].

Chemistry and Drug Discovery

In chemistry and drug discovery, generative models help to:

- Propose novel antibiotics and therapeutics with desired efficacy and safety profiles [2].
- Optimize molecules for multiple objectives (binding, solubility, toxicity, synthesizability) [3, 11].
- Accelerate clinical progress: AI-assisted drug candidates achieved Phase I success rates of nearly 90%, compared with industry averages of 40–65% [13].

Life and Environmental Sciences

- **Protein Engineering:** AlphaFold 3 predicts the structure and interactions of proteins, DNA, RNA, ligands, and ions with unprecedented accuracy, achieving at least 50% improvement over existing methods [5].
- **Genomics and Systems Biology:** Modeling sequence–function relationships and generating candidate variants for experimental follow-up [1].
- **Medical AI:** Supporting risk prediction, trial design, and treatment-effect modeling when combined with causal and statistical frameworks [1, 12].

Climate and Earth System Science

- **Weather Forecasting:** GenCast, a diffusion-based probabilistic model, outperforms the world's leading operational forecast (ECMWF ENS) in 97% of scenarios, delivering 15-day forecasts in 8 minutes [9, 14].
- **Earth System Foundations:** Large foundation models that integrate atmosphere, ocean, land, and cryosphere processes, including Google's Aurora and WeatherNext models [10].
- **Surrogate Modeling:** Physics-informed and data-driven surrogates that emulate expensive numerical models at a fraction of the computational cost [8–10, 23].

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Mathematical Foundations and Methods

Generative AI is also driving progress in the mathematics and methodology of **scientific machine learning**:

- **Data-Driven Dynamical Systems:** Learning governing dynamics and control laws directly from measurements or simulations [12].
- **Physics-Informed Neural Networks (PINNs):** Incorporating partial differential equations and other physical constraints directly into neural network training [24, 25].
- **Uncertainty Quantification:** Developing methods to quantify confidence and propagate uncertainty through AI-based scientific pipelines [25].
- **Formal Mathematical Reasoning:** AlphaProof combines reinforcement learning with formal theorem proving in Lean to generate verifiable mathematical proofs [21].

* * *

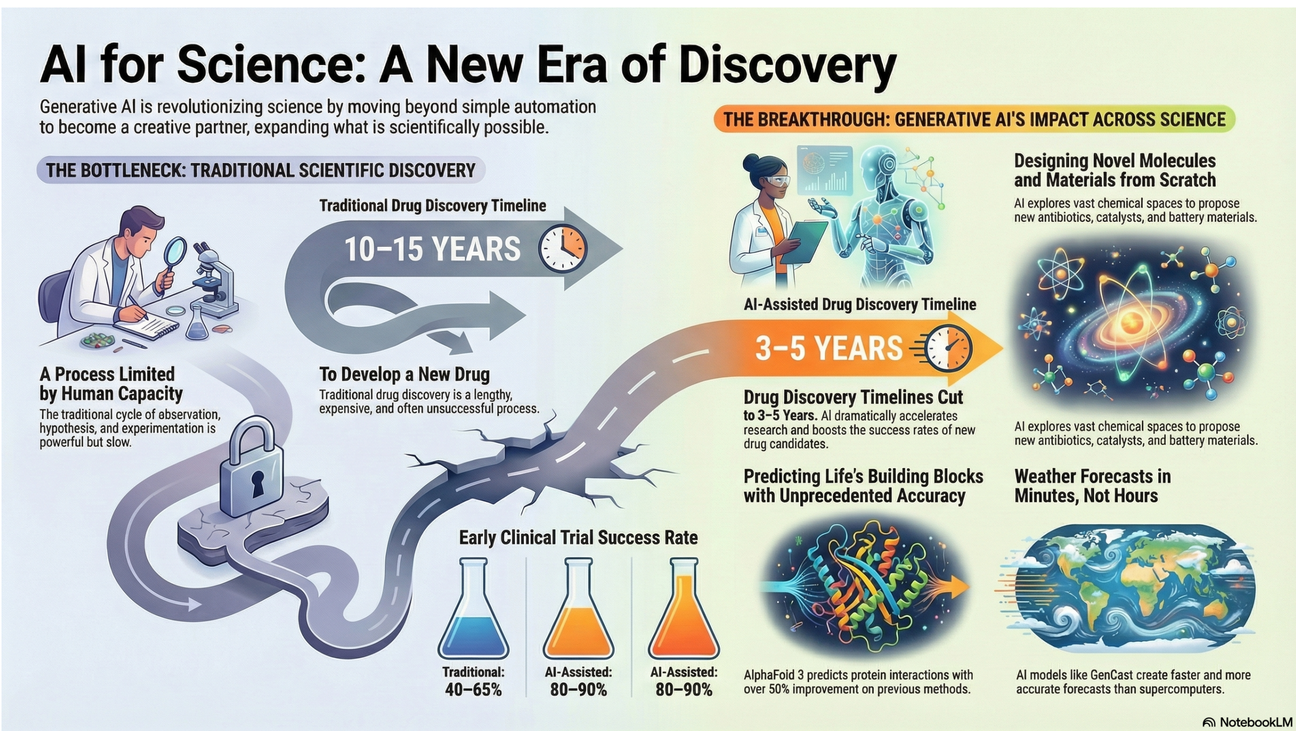

Figure 1. Chapter1.png

Figure 1.1. AI for Science: A New Era of Discovery. Traditional scientific discovery (left) is limited by human capacity, with drug development taking 10–15 years and early clinical trial success rates of 40–65%. Generative AI (right) is transforming this landscape by reducing drug discovery timelines to 3–5 years, improving clinical trial success rates to 80–90%, enabling novel molecule and materials design, predicting protein structures with unprecedented accuracy (AlphaFold 3), and accelerating weather forecasts from hours to minutes (GenCast). (Created by Google NotebookLM)

Cross-Cutting Capabilities

Several techniques and ideas cut across scientific domains:

- **Physics-Informed Models:** Embedding conservation laws, symmetries, and PDEs in neural architectures to ensure physically consistent predictions [24, 25].
- **Latent-Variable and Generative Representations:** Using VAEs, GANs, and diffusion models to learn compact representations of high-dimensional systems [16–20].
- **Multimodal Integration:** Combining text, images, time series, and spatial fields to build holistic models of complex phenomena [1, 6].
- **Multi-Agent AI Systems:** Frameworks like Google's AI co-scientist use specialized agents for hypothesis generation, debate, and evolution [7].

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A New Scientific Partner

Generative AI is *not* replacing scientists—it is becoming a **new kind of collaborator** [1]. It can:

- Expand the hypothesis space far beyond what any individual or team could enumerate
- Accelerate cycles of modeling, simulation, and experiment
- Enable researchers with limited computational or institutional resources to use state-of-the-art tools
- Bridge previously separate fields by revealing shared patterns in their data and models
- Recapitulate discoveries in days that previously required years of iterative research [7]

The most impactful discoveries are likely to emerge where **human insight** and **machine-generated ideas** interact in a tight loop.

* * *

The Path Forward

The rest of this book will help you:

- Understand the **core architectures** (transformers, diffusion models, scientific ML) [15–18, 24, 25].
- Adapt and fine-tune models to your own **scientific domain** [2–5, 11].
- Evaluate and validate models with appropriate **statistical and physical checks** [12, 23, 25].
- Design human–AI workflows in which AI accelerates, rather than distorts, the scientific process [1].

The revolution is already underway. In my view, **scientists who understand and apply generative AI will stand at the forefront of discovery in every field** [1].

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References and Further Readings

1. **Wang, H.**, Fu, T., Du, Y., Gao, W., Huang, K., Liu, Z., *et al.* (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620(7972), 47–60. https://doi.org/10.1038/s41586-023-06221-2
2. **Stokes, J. M.**, Yang, K., Swanson, K., Jin, W., Cubillos-Ruiz, A., Donghia, N. M., *et al.* (2020). A deep learning approach to antibiotic discovery. *Cell*, 180(4), 688–702. https://doi.org/10.1016/j.cell.2020.01.021
3. **Gómez-Bombarelli, R.**, Wei, J. N., Duvenaud, D., Hernández-Lobato, J. M., Sánchez-Lengeling, B., Sheberla, D., *et al.* (2018). Automatic chemical design using a data-driven continuous representation of molecules. *ACS Central Science*, 4(2), 268–276. https://doi.org/10.1021/acscentsci.7b00572
4. **Jumper, J.**, Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., *et al.* (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2
5. **Abramson, J.**, Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., *et al.* (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630(8016), 493–500. https://doi.org/10.1038/s41586-024-07487-w
6. **Brown, T.**, Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., *et al.* (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877–1901. https://arxiv.org/abs/2005.14165
7. **Gottweis, J.**, Weng, W.-H., Daryin, A., Liu, Y., Ceze, L., Natarajan, V., *et al.* (2025). Towards an AI co-scientist. *arXiv preprint* arXiv:2502.18864. https://arxiv.org/abs/2502.18864
8. **Reichstein, M.**, Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., & Prabhat. (2019). Deep learning and process understanding for data-driven Earth system science. *Nature*, 566(7743), 195–204. https://doi.org/10.1038/s41586-019-0912-1

9. **Price, I.**, Sanchez-Gonzalez, A., Alet, F., Andersson, T. R., El-Kadi, A., Masters, D., *et al.* (2024). Probabilistic weather forecasting with machine learning. *Nature*, 636, 84–90. https://doi.org/10.1038/s41586-024-08252-9
10. **Bodnar, C.**, Bruinsma, W. P., Lucic, A., Stanley, M., Allen, A., Brandstetter, J., *et al.* (2024). Aurora: A foundation model of the atmosphere. *arXiv preprint* arXiv:2405.13063. https://arxiv.org/abs/2405.13063
11. **Angello, N. H.**, Friday, D. M., Hwang, C., Yi, S., Cheng, A. H., Torres-Flores, T. C., *et al.* (2024). Closed-loop transfer enables artificial intelligence to yield chemical knowledge. *Nature*, 633(8029), 351–358. https://doi.org/10.1038/s41586-024-07021-y
12. **Brunton, S. L.**, & Kutz, J. N. (2022). *Data-driven science and engineering: Machine learning, dynamical systems, and control* (2nd ed.). Cambridge University Press. https://doi.org/10.1017/9781108380690
13. **AI in Drug Discovery** (2024). Industry analysis reports indicate AI-assisted drug candidates achieving Phase I success rates of nearly 90%, compared with industry averages of 40–65%. *Drug Discovery News*, October 2025. https://pubmed.ncbi.nlm.nih.gov/38692505
14. **Google DeepMind** (2024). GenCast predicts weather and the risks of extreme conditions with state-of-the-art accuracy. https://deepmind.google/blog/gencast-predicts-weather-and-the-risks-of-extreme-conditions-with-sota-accuracy/
15. **Vaswani, A.**, Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., *et al.* (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. https://arxiv.org/abs/1706.03762
16. **Ho, J.**, Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851. https://arxiv.org/abs/2006.11239
17. **Dhariwal, P.**, & Nichol, A. (2021). Diffusion models beat GANs on image synthesis. *Advances in Neural Information Processing Systems*, 34, 8780–8794. https://arxiv.org/abs/2105.05233
18. **Song, Y.**, Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2021). Score-based generative modeling through stochastic differential equations. *Proceedings of ICLR*. https://arxiv.org/abs/2011.13456
19. **Kingma, D. P.**, & Welling, M. (2014). Auto-encoding variational Bayes. *Proceedings of ICLR*. https://arxiv.org/abs/1312.6114
20. **Goodfellow, I.**, Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., *et al.* (2014). Generative adversarial networks. *Advances in Neural Information Processing Systems*, 27. https://arxiv.org/abs/1406.2661
21. **Google DeepMind** (2024). AI achieves silver-medal standard solving International Mathematical Olympiad problems. https://deepmind.google/blog/ai-solves-imo-problems-at-silver-medal-level/
22. **Google DeepMind** (2025). Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad. https://deepmind.google/blog/advanced-version-of-gemini-with-deep-think-officially-achieves-gold-medal-standard-at-the-international-mathematical-olympiad/

23. **Carleo, G.**, Cirac, I., Cranmer, K., Daudet, L., Schuld, M., Tishby, N., *et al.* (2019). Machine learning and the physical sciences. *Reviews of Modern Physics*, 91(4), 045002. https://doi.org/10.1103/RevModPhys.91.045002
24. **Raissi, M.**, Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378, 686–707. https://doi.org/10.1016/j.jcp.2018.10.045
25. **Karniadakis, G. E.**, Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., & Yang, L. (2021). Physics-informed machine learning. *Nature Reviews Physics*, 3(6), 422–440. https://doi.org/10.1038/s42254-021-00314-5

Next → Chapter 2: introduces the core architectures powering generative AI.

Chapter 2: Generative AI Fundamentals

Introduction: The Building Blocks of Generation

To harness generative AI for scientific discovery, we must understand how these models work—not as black boxes, but as computational systems with clear mathematical foundations and architectural principles. This chapter demystifies the core technologies powering modern generative AI: transformers, diffusion models, flow matching, and variational approaches using plain English. We'll explore their mechanisms, understand when to use each, and see how they can be adapted for scientific applications.

While the mathematics can become complex, our focus is on building intuition. Scientists don't need to be machine learning experts to use these tools effectively, but understanding the fundamentals will help you choose the right model architecture, diagnose failures, adapt pre-trained models, collaborate with computational researchers, and evaluate limitations.

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The Three Pillars of Generative AI

Modern generative AI rests on three major architectural families:

Architecture	Core Mechanism	Best For	Example Applications
Transformers & LLMs	Self-attention over sequences	Text, code, sequences	Literature synthesis, protein sequences, code generation
Diffusion & Flow Models	Iterative denoising / flow matching	Structured outputs, images	Molecular structures, protein folding, climate data
VAEs & GANs	Latent space learning	Data generation, interpolation	Synthetic data, anomaly detection, compression

Architecture	Core Mechanism	Best For	Example Applications

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Part I: Transformers and Large Language Models

The Attention Revolution

Traditional neural networks process sequences step-by-step. **Transformers** changed this with attention: let every element directly attend to every other element simultaneously [1]. The breakthrough "Attention Is All You Need" paper introduced self-attention mechanisms that model long-range dependencies in parallel, eliminating the sequential bottleneck of recurrent networks.

The Attention Mechanism

Attention computes three quantities for each sequence element [1]:

Component	Role	Intuition
Query (Q)	What am I looking for?	The question each element asks
Key (K)	What do I contain?	How each element describes itself
Value (V)	What do I contribute?	The information each element provides

Attention formula:

```
Attention(Q, K, V) = softmax(Q·K^T / √d_k) · V
```

Here, "softmax" turns raw similarity scores into a meaningful probability distribution that tells the model how much attention to pay to each token.

Why This Works for Science:

- **Protein Sequences**: Connect distant amino acids that interact in 3D
- **Scientific Literature**: Link concepts mentioned paragraphs apart
- **Chemical Reactions**: Identify relationships between reactants and products

The Transformer Architecture

The original encoder-decoder architecture [1] consists of stacked attention and feed-forward layers:

```
Input → Embedding → Positional Encoding
    ↓
[ Multi-Head Attention → Add & Norm
    ↓
  Feed-Forward → Add & Norm ] × N layers
    ↓
Output Probabilities
```

Key Components:

Component	Purpose	Scientific Benefit
Multi-Head Attention	Learn different patterns	Capture multiple relationship types
Residual Connections	Enable deep networks	Scale to billions of parameters
Layer Normalization	Stabilize training	Faster convergence
Positional Encoding	Track sequence order	Understand structure (DNA direction, time series)

From Transformers to Large Language Models

1. Pre-Training: Learning General Patterns

Objective: Predict next token given context.

```
Input:  "The protein binds to the"
Target: "receptor"
```

The GPT series demonstrated the power of autoregressive language modeling at scale [2, 3], while BERT showed bidirectional pre-training for understanding tasks [4]. More recent models like LLaMA [5], Gemini [6], and Claude push the boundaries with trillions of tokens and multimodal capabilities.

Scientific Pre-Trained Models:

- **SciBERT** [7]: Scientific papers
- **ESM-2** [8]: Protein sequences (750M sequences)
- **ESM-3** [9]: Multimodal protein model (sequence, structure, function) with 98B parameters (2024/2025)
- **Llama 3.1/3.2/3.3** [10]: Open foundation models with 128K context (2024)

2. Fine-Tuning: Domain Specialization

Full fine-tuning updates all model parameters, but parameter-efficient methods like LoRA [11] enable adaptation with minimal computational cost by learning low-rank updates to weight matrices.

Method	Trainable Params	Data Needed	Use Case
Full Fine-Tuning	100%	10K-100K	Maximum adaptation
LoRA [11]	0.1-1%	100-10K	Limited compute
Adapters [12]	1-5%	1K-10K	Task-specific layers
QLoRA [13]	0.1-1%	100-10K	Quantized + LoRA (fine-tune 70B on consumer GPU)

3. Prompting: Zero-Shot and Few-Shot

Large language models demonstrate remarkable few-shot learning capabilities [3], enabling scientific applications without task-specific training.

Zero-Shot:

```
Summarize this oceanography paper: [text]
```

Few-Shot:

```
Examples:
SMILES: CCO → Name: Ethanol
SMILES: CC(C)O → Name: Propan-2-ol

SMILES: CCCO → Name: ?
```

Reasoning Models: A New Paradigm (2024–2025)

A significant development in 2024–2025 is the emergence of **reasoning models** that "think before they respond" [14, 15]. Unlike standard LLMs that generate answers in a single pass, reasoning models like OpenAI's o1/o3 series and Gemini Deep Think produce an internal chain of thought before responding.

Key Characteristics:

- Extended reasoning time for complex problems
- Multi-step planning and self-verification
- Superior performance on math, coding, and scientific reasoning
- Configurable "reasoning effort" levels

Scientific Applications:

- Complex mathematical proofs
- Multi-step experimental design
- Code debugging and algorithm implementation
- Hypothesis evaluation requiring logical chains

Limitations for Science

Challenge	Impact	Mitigation
Hallucination	False information	RAG [16], verification, reasoning models
Lack of Uncertainty	Overconfidence	Ensemble methods [17]
Data Cutoff	Outdated knowledge	Fine-tuning, RAG [16]
Context Limits	Long document handling	Extended context models (128K+ tokens)

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Part II: Diffusion Models and Flow Matching

The Denoising Paradigm

Diffusion models learn to generate by reversing a noise-addition process [18, 19]. The foundational work by Sohl-Dickstein et al. [18] established the thermodynamic interpretation, while Ho et al. [19] simplified training with the denoising objective.

Forward Process: Adding Noise

$$x_0 \text{ (real data)} \rightarrow x_1 \rightarrow x_2 \rightarrow \ldots \rightarrow x_t \text{ (pure noise)}$$

At each step: $x_t = \sqrt{(1-\beta_t)} \cdot x_{t-1} + \sqrt{\beta_t} \cdot \varepsilon$ where $\varepsilon \sim N(0,I)$

Reverse Process: Learning to Denoise

Train a network to predict the noise [19]:

$$\text{Loss} = E[||\varepsilon - \varepsilon_{pre□}(x_t, t)||^2]$$

Generation: Sampling

1. Start with noise: $x_t \sim N(0, I)$
2. Iteratively denoise: $x_t \rightarrow x_{t-1} \rightarrow \ldots \rightarrow x_0$
3. Result: novel sample

Flow Matching: A Powerful Alternative (2023–2025)

Flow Matching [20] has emerged as a powerful and efficient alternative to diffusion-based generative modeling, with growing interest in scientific applications [21]. Rather than learning to reverse a noising process, flow matching learns a velocity field that transports samples from noise to data along continuous paths.

Key Advantages:

- **Straighter trajectories**: Fewer sampling steps required
- **Stable training**: Simpler objectives, less hyperparameter tuning
- **Faster inference**: Often 2-4x speedup over diffusion
- **Flexible conditioning**: Natural incorporation of constraints

Flow Matching in Biology (2024–2025):

- Molecule generation (NeurIPS 2023) [21]
- Protein backbone generation with SE(3)-equivariant flows (ICLR 2024) [21]
- Antibody design with IgFlow and dyAb [21]
- Biological sequence and peptide generation (ICML 2024) [21]

Why Diffusion/Flow Works for Science

Score-based generative modeling [22] provides a continuous-time perspective, while empirical results show diffusion models produce higher-quality samples than GANs [23].

Feature	Scientific Benefit
High Quality	Realistic structures
Stable Training	Easier than GANs [23]
Interpretable	Visualize generation
Conditional	Incorporate constraints [24]
Uncertainty	Multiple samples

Applications:

- Molecular conformations with valency constraints [25]
- Protein structures respecting physics [26]
- Climate data filling gaps while conserving energy
- High-resolution image synthesis with latent diffusion [27]
- Probabilistic weather forecasting with GenCast [28]

Conditional Diffusion

Generate data with specific properties using classifier guidance or classifier-free guidance [24]:

$p(x_{t-1} \mid x_t, \text{condition})$

Conditioning Methods:

Method	Example
Classifier Guidance [23]	"Molecule binds to protein X"
Classifier-Free [24]	Train with/without conditions
Inpainting	Fill missing climate data

* * *

Part III: VAEs and GANs

Variational Autoencoders (VAEs)

VAEs learn probabilistic latent representations through variational inference [29]:

Architecture:

```
Encoder: x → [μ(x), σ(x)]
Sample: z ~ N(μ, σ²)
Decoder: z → x̂
```

Loss:

```
Total = Reconstruction + KL_Divergence
```

Scientific Uses:

- **Chemistry**: Interpolate between molecules [30]
- **Materials**: Optimize in latent space
- **Genomics**: Compress gene expression data

Generative Adversarial Networks (GANs)

GANs introduced adversarial training between generator and discriminator [31]:

Two Networks in Competition:

Network	Role	Goal
Generator	Create fakes	Fool discriminator
Discriminator	Judge real/fake	Detect fakes

Challenges:

- Mode collapse (limited diversity)
- Training instability
- Difficult evaluation

Scientific Uses:

- Data augmentation
- Super-resolution
- Image-to-image translation
- Molecular graph generation [32]

Comparison

Criterion	VAE	GAN	Diffusion	Flow Matching
Quality	Good	Excellent	Excellent	Excellent
Stability	Stable	Unstable	Stable	Very Stable
Diversity	Good	Poor	Excellent	Excellent
Speed	Fast	Fast	Slow	Moderate
Latent Space	Interpretable	Opaque	Implicit	Implicit

* * *

Part IV: Pre-Training and Fine-Tuning

The Transfer Learning Pipeline

```
Pre-Training (general data, millions of examples)
    ↓
Fine-Tuning (domain data, thousands of examples)
    ↓
Prompting (task-specific, zero examples)
```

Foundation models [33] trained on massive corpora can be adapted to downstream scientific tasks with limited data.

Parameter-Efficient Fine-Tuning (PEFT)

LoRA Example [11]:

```python
from peft import LoraConfig, get_peft_model

config = LoraConfig(
    r=16,  # Adaptation rank
    lora_alpha=32,
    target_modules=["q_proj", "v_proj"]
)

model = get_peft_model(base_model, config)
# Now only 0.5% of parameters are trainable!
```

QLoRA [13] combines quantization with LoRA, enabling fine-tuning of 70B models on consumer GPUs—a game-changer for scientific labs with limited compute budgets.

Benefits:

- Train on single GPU
- Fast iteration
- Multiple task-specific versions

Open vs. Closed Models for Science

The 2024–2025 landscape offers scientists unprecedented choice:

Model Family	Access	Parameters	Best For
Llama 3.1 / 3.2 / 3.3	Open weights	1B–405B (text); 11B / 90B (vision variants)	Local deployment, fine-tuning, on-prem and private customization
OpenAI GPT-4/4o/5.1/5.2	API (and ChatGPT)	Not disclosed	Advanced reasoning, multimodal understanding, coding, tool-using agents
Gemini 2.0/3.0 (Pro, Flash)	API (Gemini API, Vertex AI)	Not disclosed	Multimodal tasks, very long context (up to ~1M tokens on supported models)
Claude (Claude 3.5, Sonnet/Opus 4.5)	API	Not disclosed	Deep analysis, coding, safety-oriented assistants, long-context workflows
**ESM-3 **	API + open weights (ESM3-open)	1.4B (open); up to ~98B (largest)	Protein modeling, structure/function prediction, generative protein design

* * *

Part V: Mathematical Foundations

Low-Rank Approximation

Large models use low-rank structures for efficiency [11].

Matrix Factorization:

```
W ∈ R^(m×n) ≈ A·B where A ∈ R^(m×r), B ∈ R^(r×n), r << min(m,n)
```

Applications:

- LoRA adaptation [11]
- Model compression
- Fast inference

Quantization

Reduce precision for efficiency [34, 35]:

Precision	Bits	Range	Use Case
FP32	32	Full	Training
FP16/BF16	16	Mixed	Fast training
INT8	8	Limited	Inference
INT4	4	Very limited	Edge deployment, QLoRA

LLM.int8() [34] and GPTQ [35] enable accurate quantization without retraining.

Benefits:

- 4× smaller models (FP32 → INT8)
- 2-4× faster inference
- Run large models on consumer GPUs

Optimization Theory

Key Algorithms:

Optimizer	Mechanism	When to Use
SGD	Basic gradient descent	Small models, well-understood problems
Adam [36]	Adaptive learning rates	Default choice, most robust
AdamW [37]	Adam + weight decay	Large language models

Learning Rate Schedules:

```
Warmup → Constant → Decay
```

- **Warmup**: Start small, gradually increase (stabilize training)
- **Constant**: Maintain learning rate (main training)
- **Decay**: Reduce toward end (fine-tune solution)

* * *

Part VI: Types of Generative AI by Modality

Text Generation

Capabilities:

- Scientific writing
- Code generation
- Literature synthesis
- Hypothesis generation

Models: GPT-4o/4.5/o1/o3 [14, 15], Llama 3.1/3.2/3.3/4 [10], Gemini 2.0/3.0 [6], Claude.

Image Generation

Capabilities:

- Synthetic microscopy
- Medical imaging
- Scientific visualization
- Data augmentation

Models: Stable Diffusion 3/3.5 [38], SDXL [27], DALL-E 3

Molecular Generation

Capabilities:

- Drug discovery
- Material design
- Reaction prediction

Representations:

- SMILES strings (text-like)
- Molecular graphs [32]
- 3D conformations [25]

Models: MolGAN [32], Diffusion-based generators [25], VAEs [30], Flow matching models [21]

Protein Generation

Capabilities:

- Structure prediction (AlphaFold 3 [39], ESMFold [8])
- Sequence design
- Function prediction
- Multimodal generation (ESM-3) [9]

Models: ESM-2 [8], ESM-3 [9], ProtGPT2 [40], RFdiffusion [26]

2024–2025 Highlight: ESM-3 [9] ESM-3, published in *Science* (January 2025), is a 98B parameter multimodal generative model that reasons over protein sequence, structure, and function simultaneously. It generated a novel green fluorescent protein (GFP) with only 58% sequence identity to known GFPs—equivalent to simulating 500 million years of evolution.

Graph Generation

Capabilities:

- Molecular graphs
- Protein-protein interaction networks
- Knowledge graphs

Models: GraphRNN, MolGAN [32], GraphVAE

Multimodal Generation

Capabilities:

- Text → Image (CLIP [41] + diffusion)
- Text → Molecule (MolT5)
- Image → Text (scientific figure captioning)
- Cross-modal retrieval

Models: CLIP [41], GPT-4o/4V [15], Gemini 2.0 [6]

* * *

Design Principles for Scientific Applications

1. Incorporate Domain Knowledge

Physics-Informed Neural Networks (PINNs) [42]: Embed differential equations in loss function:

```
Loss = Data_Loss + λ·Physics_Loss

Physics_Loss = ||∂u/∂t - α∇²u||²  (heat equation)
```

Benefits:

- Better generalization
- Physical consistency
- Data efficiency

2. Quantify Uncertainty

Methods:

- **Ensemble** [17]: Train multiple models, measure spread
- **Bayesian** [43]: Maintain distributions over parameters
- **Conformal** [44]: Finite-sample coverage guarantees

Why Essential: Scientific decisions have consequences—we must know when models are uncertain.

3. Ensure Reproducibility

Checklist:

- Set random seeds
- Log hyperparameters
- Version control data and code
- Document environment
- Share trained models

4. Validate Rigorously

Validation Type	Purpose
Hold-out Test	Independent performance
Cross-validation	Robust estimates
Domain Expert	Scientific plausibility
Physical Constraints	Obey natural laws
Out-of-distribution	Generalization limits

* * *

Practical Considerations

Model Selection Guide

Your Goal	Recommended Architecture	Rationale
Generate scientific text	Transformer LLM [2, 3, 10]	Excellent at language
Complex reasoning/math	Reasoning models (o1/o3) [14, 15]	Multi-step verification
Design molecules	Diffusion [25] or Flow Matching [21]	High-quality structures

Your Goal	Recommended Architecture	Rationale
Predict protein structure	ESM-3 [9], AlphaFold 3 [39]	State-of-the-art specialized models
Fill missing data	Conditional diffusion [24]	Respects constraints
Synthetic data augmentation	GAN [31] or VAE [29]	Fast generation
Explore design space	VAE [29]	Interpretable latent space

Computational Requirements

Task	Typical Resources	Alternative
Fine-tune small LLM	Single GPU, 1-2 days	Use LoRA/QLoRA [11, 13] for efficiency
Train diffusion model	4-8 GPUs, 3-7 days	Use pre-trained models [27]
Inference (LLM)	1 GPU or CPU	Quantization [34, 35] for speed
Protein structure	1 GPU, minutes	Use hosted APIs

Common Pitfalls

Pitfall	Consequence	Solution
Overfitting	Poor generalization	Regularization, more data
Underfitting	Poor performance	Larger model, more training
Data leakage	Inflated metrics	Careful splitting
Ignoring uncertainty	Overconfident predictions	Uncertainty quantification [17, 43, 44]
Black-box use	Unexplainable failures	Understand fundamentals

* * *

Summary

This chapter introduced the core architectures powering generative AI:

Transformers [1] excel at sequential data through attention mechanisms, enabling scientific text generation [2, 3, 10], code synthesis, and protein language models [8, 9]. The emergence of **reasoning models** [14, 15] in 2024–2025 represents a paradigm shift for complex scientific problem-solving.

Diffusion models [18, 19, 22] and **flow matching** [20, 21] generate high-quality structured outputs by learning to reverse noise or transport samples along continuous paths, ideal for molecular design [25], protein structures [26], and climate data reconstruction [28].

VAEs [29] and **GANs** [31] provide complementary approaches for data generation, exploration, and augmentation.

Transfer learning [33]—pre-training followed by fine-tuning [11, 12, 13]—allows scientists to leverage massive general models with modest domain-specific data. Open models like Llama 3/4 [10] democratize access for scientific labs.

Mathematical foundations like low-rank approximation [11], quantization [34, 35], and optimization theory [36, 37] enable efficient training and deployment.

Understanding these fundamentals empowers you to:

- Choose appropriate models for your scientific problems
- Adapt existing models to new domains
- Diagnose and fix failures
- Collaborate effectively with AI researchers
- Push the boundaries of what's possible in your field

* * *

References

1. **Vaswani, A.**, Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., *et al.* (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. https://arxiv.org/abs/1706.03762
2. **Radford, A.**, Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners (GPT-2). *OpenAI Blog*. https://openai.com/research/better-language-models

3. **Brown, T.**, Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., *et al.* (2020). Language models are few-shot learners (GPT-3). *Advances in Neural Information Processing Systems*, 33, 1877–1901. https://arxiv.org/abs/2005.14165
4. **Devlin, J.**, Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, 4171–4186. https://arxiv.org/abs/1810.04805
5. **Touvron, H.**, Lavril, T., Izacard, G., Martinet, X., Lachaux, M. A., Lacroix, T., *et al.* (2023). LLaMA: Open and efficient foundation language models. *arXiv preprint* arXiv:2302.13971. https://arxiv.org/abs/2302.13971
6. **Gemini Team, Google**. (2024). Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint* arXiv:2403.05530. https://arxiv.org/abs/2403.05530
7. **Beltagy, I.**, Lo, K., & Cohan, A. (2019). SciBERT: A pretrained language model for scientific text. *Proceedings of EMNLP-IJCNLP*, 3615–3620. https://arxiv.org/abs/1903.10676
8. **Lin, Z.**, Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., *et al.* (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model (ESM-2/ESMFold). *Science*, 379(6637), 1123–1130. https://doi.org/10.1126/science.ade2574
9. **Hayes, T.**, Rao, R., Akin, H., Sofroniew, N. J., Oktay, D., Lin, Z., *et al.* (2025). Simulating 500 million years of evolution with a language model (ESM-3). *Science*, 387(6736), 850–858. https://doi.org/10.1126/science.ads0018
10. **Meta AI**. (2024). Llama 3.1, 3.2, and 3.3: The most capable openly available LLMs. *Meta AI Blog*. https://ai.meta.com/blog/meta-llama-3-1/
11. **Hu, E. J.**, Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., *et al.* (2022). LoRA: Low-rank adaptation of large language models. *Proceedings of ICLR*. https://arxiv.org/abs/2106.09685
12. **Houlsby, N.**, Giurgiu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., *et al.* (2019). Parameter-efficient transfer learning for NLP. *Proceedings of ICML*, 2790–2799. https://arxiv.org/abs/1902.00751
13. **Dettmers, T.**, Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2024). QLoRA: Efficient finetuning of quantized LLMs. *Proceedings of NeurIPS*, 36. https://arxiv.org/abs/2305.14314
14. **OpenAI**. (2024). Introducing OpenAI o1. *OpenAI Blog*. https://openai.com/index/introducing-openai-o1-preview/
15. **OpenAI**. (2024). GPT-4o and reasoning models. *OpenAI Platform*. https://platform.openai.com/docs/models
16. **Lewis, P.**, Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., *et al.* (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Proceedings of NeurIPS*, 33, 9459–9474. https://arxiv.org/abs/2005.11401
17. **Lakshminarayanan, B.**, Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. *Proceedings of NeurIPS*, 30. https://arxiv.org/abs/1612.01474

18. **Sohl-Dickstein, J.**, Weiss, E., Maheswaranathan, N., & Ganguli, S. (2015). Deep unsupervised learning using nonequilibrium thermodynamics. *Proceedings of ICML*, 2256–2265. https://arxiv.org/abs/1503.03585
19. **Ho, J.**, Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851. https://arxiv.org/abs/2006.11239
20. **Lipman, Y.**, Chen, R. T. Q., Ben-Hamu, H., Nickel, M., & Le, M. (2023). Flow matching for generative modeling. *Proceedings of ICLR*. https://arxiv.org/abs/2210.02747
21. **Li, Z.**, Zeng, Z., Lin, X., Fang, F., Qu, Y., Xu, Z., *et al.* (2025). Flow matching meets biology and life science: A survey. *arXiv preprint* arXiv:2507.17731. https://arxiv.org/abs/2507.17731
22. **Song, Y.**, Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2021). Score-based generative modeling through stochastic differential equations. *Proceedings of ICLR*. https://arxiv.org/abs/2011.13456
23. **Dhariwal, P.**, & Nichol, A. (2021). Diffusion models beat GANs on image synthesis. *Advances in Neural Information Processing Systems*, 34, 8780–8794. https://arxiv.org/abs/2105.05233
24. **Ho, J.**, & Salimans, T. (2022). Classifier-free diffusion guidance. *NeurIPS Workshop on Score-Based Methods*. https://arxiv.org/abs/2207.12598
25. **Hoogeboom, E.**, Satorras, V. G., Vignac, C., & Welling, M. (2022). Equivariant diffusion for molecule generation in 3D. *Proceedings of ICML*, 8867–8887. https://arxiv.org/abs/2203.17003
26. **Watson, J. L.**, Juergens, D., Bennett, N. R., Trippe, B. L., Yim, J., Eisenach, H. E., *et al.* (2023). De novo design of protein structure and function with RFdiffusion. *Nature*, 620(7976), 1089–1100. https://doi.org/10.1038/s41586-023-06415-8
27. **Rombach, R.**, Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models (Stable Diffusion). *Proceedings of CVPR*, 10684–10695. https://arxiv.org/abs/2112.10752
28. **Price, I.**, Sanchez-Gonzalez, A., Alet, F., Andersson, T. R., El-Kadi, A., Masters, D., *et al.* (2024). Probabilistic weather forecasting with machine learning (GenCast). *Nature*, 636, 84–90. https://doi.org/10.1038/s41586-024-08252-9
29. **Kingma, D. P.**, & Welling, M. (2014). Auto-encoding variational Bayes. *Proceedings of ICLR*. https://arxiv.org/abs/1312.6114
30. **Gómez-Bombarelli, R.**, Wei, J. N., Duvenaud, D., Hernández-Lobato, J. M., Sánchez-Lengeling, B., Sheberla, D., *et al.* (2018). Automatic chemical design using a data-driven continuous representation of molecules. *ACS Central Science*, 4(2), 268–276. https://doi.org/10.1021/acscentsci.7b00572
31. **Goodfellow, I.**, Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., *et al.* (2014). Generative adversarial networks. *Advances in Neural Information Processing Systems*, 27. https://arxiv.org/abs/1406.2661
32. **De Cao, N.**, & Kipf, T. (2018). MolGAN: An implicit generative model for small molecular graphs. *ICML Workshop on Theoretical Foundations and Applications of Deep Generative Models*. https://arxiv.org/abs/1805.11973

33. **Bommasani, R.**, Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., *et al.* (2021). On the opportunities and risks of foundation models. *arXiv preprint* arXiv:2108.07258. https://arxiv.org/abs/2108.07258
34. **Dettmers, T.**, Lewis, M., Belkada, Y., & Zettlemoyer, L. (2022). LLM.int8(): 8-bit matrix multiplication for transformers at scale. *Proceedings of NeurIPS*, 35, 30318–30332. https://arxiv.org/abs/2208.07339
35. **Frantar, E.**, Ashkboos, S., Hoefler, T., & Alistarh, D. (2023). GPTQ: Accurate post-training quantization for generative pre-trained transformers. *Proceedings of ICLR*. https://arxiv.org/abs/2210.17323
36. **Kingma, D. P.**, & Ba, J. (2015). Adam: A method for stochastic optimization. *Proceedings of ICLR*. https://arxiv.org/abs/1412.6980
37. **Loshchilov, I.**, & Hutter, F. (2019). Decoupled weight decay regularization (AdamW). *Proceedings of ICLR*. https://arxiv.org/abs/1711.05101
38. **Stability AI**. (2024). Stable Diffusion 3: Scaling rectified flow transformers for high-resolution image synthesis. https://stability.ai/stable-image
39. **Abramson, J.**, Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., *et al.* (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630(8016), 493–500. https://doi.org/10.1038/s41586-024-07487-w
40. **Ferruz, N.**, Schmidt, S., & Höcker, B. (2022). ProtGPT2 is a deep unsupervised language model for protein design. *Nature Communications*, 13(1), 4348. https://doi.org/10.1038/s41467-022-32007-7
41. **Radford, A.**, Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., *et al.* (2021). Learning transferable visual models from natural language supervision (CLIP). *Proceedings of ICML*, 8748–8763. https://arxiv.org/abs/2103.00020
42. **Raissi, M.**, Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378, 686–707. https://doi.org/10.1016/j.jcp.2018.10.045
43. **Gal, Y.**, & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. *Proceedings of ICML*, 1050–1059. https://arxiv.org/abs/1506.02142
44. **Angelopoulos, A. N.**, & Bates, S. (2021). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint* arXiv:2107.07511. https://arxiv.org/abs/2107.07511

Additional Resources

Textbooks

- **Goodfellow, I.**, Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. http://www.deeplearningbook.org

- **Prince, S. J. D.** (2023). *Understanding Deep Learning*. MIT Press. http://udlbook.github.io/udlbook/
- **Tunstall, L.**, von Werra, L., & Wolf, T. (2022). *Natural Language Processing with Transformers*. O'Reilly Media.
- **Alammar, J.**, & van den Oord, A. (2024). *Generative Deep Learning* (2nd ed.). O'Reilly Media. https://www.oreilly.com/library/view/generative-deep-learning/9781098134174/
- **Raschka, S.** (2024). *Build a Large Language Model (From Scratch)*. Manning. https://www.manning.com/books/build-a-large-language-model-from-scratch
- **Holderrieth, P.**, & Erives, E. (2025). *Introduction to Flow Matching and Diffusion Models*. MIT Course 6.S184. https://diffusion.csail.mit.edu/
- **Liu, J.P.** (2025). *How to Build and Fine-tune a Small Language Model*. Leanpub. https://leanpub.com/howtobuildandfine-tuneasmalllanguagemodel

Key Review Papers

- **Yang, L.**, et al. (2023). Diffusion models: A comprehensive survey. *ACM Computing Surveys*, 56(4), 1–39.
- **Bommasani, R.**, et al. (2022). On the opportunities and risks of foundation models. https://arxiv.org/abs/2108.07258
- **Zhao, W. X.**, et al. (2023). A survey of large language models. *ACM Computing Surveys*. https://arxiv.org/abs/2303.18223
- **Li, Z.**, et al. (2025). Flow matching meets biology and life science: A survey. https://arxiv.org/abs/2507.17731

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This chapter provides the foundational knowledge needed to understand and apply generative AI in scientific contexts. For the latest developments, regularly check arXiv (cs.LG, cs.CL, q-bio sections) and major conference proceedings (NeurIPS, ICML, ICLR, CVPR).

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Next → Chapter 3: Scientific Data & Workflows—navigating the unique challenges of scientific datasets and integrating AI into research pipelines.

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